

Open Problems and System Cases

The paper card exposes gaps at the interfaces between model, runtime, temporal module, and task contract.

Open problems exposed by the ledger

Each gap links a missing capability to a layer and an evidence requirement.

Temporal and numerical faithfulness	Typed time operations, units, chronology guards, and end-to-end fidelity.	GTS
Temporal provenance and memory	Point-in-time retrieval, revision history, regime awareness, and replayable evidence.	GH + GTS
Interface standardization	Shared schemas for temporal workspaces, tools, traces, and validators.	GH + GTS
Safe component evolution	Regression checks and held-out temporal tests for new tools, policies, and memories.	GH
Decision-centered evaluation	Connect forecasts and diagnoses to action quality, cost, uncertainty, and review.	TSK

Focused system cases

Three cards show different contribution loci.

ChatTS	LLM side
Temporal alignment and dialogue supervision make numerical series available to language reasoning.	
TimeClaw	GTS runtime
Temporal tools, auditable trajectories, memory, and capability evolution become executable.	
ModernTSF	Infrastructure
Models, datasets, artifacts, budgets, and recoverable execution support reproducible evaluation.	

The paper card records the claim, module, task contract, and supporting evidence for each case.

Paper Card

Claim

Module

Task contract

Evidence locator